Quincy Mei

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EDUCATION

Northeastern University | Expected Graduation: May 2026 | San Jose, CA
Master of Science, Major: Computer Science

University of West Alabama | Graduated: 2023 | Livingston, AL
Bachelor of Business Administration, Major: Quantitative Finance and Economics

WORK EXPERIENCE

Software Engineer Intern | June 2024 – Aug 2024
SuperADS | Guangzhou, China

- Architected a **distributed video translation microservice** with RESTful API, Redis-based job queue, and asynchronous task processing, **reducing API latency by 20%** and enabling **500+ concurrent** requests.
- Engineered high-performance **multithreaded C++ video pipelines (OpenGL, FFmpeg)** with **AI-based** subtitle removal **(STTN, LAMA, ProPainter)**, improving processing throughput by **30%** through caching and pipeline optimization.
- Designed an **end-to-end automated subtitle localization workflow** by integrating third-party translation APIs into a custom backend, accelerating delivery time from **hours to minutes**.
- Established **CI/CD pipeline** with automated unit testing via **Google Test** framework, achieving **85%** code coverage and reducing production bugs by **40%**, improving deployment reliability.

PROJECTS

GradHunt – AI-Powered Job Search Platform | AWS, Python, Node.js

- Engineered a scalable full-stack job search platform using **React**, **Vite**, **Node.js**, **Express**, **MongoDB**, and **serverless AWS**, supporting **500+** concurrent users.
- Engineered a **scalable backend pipeline** for parallel multi-source job, normalization, fuzzy deduplication, and resume-driven search expansion, with fallback handling to maintain search relevance and resilience under partial scraper failure.
- Developed **AI-powered** matching features with **Open AI**, including structured resume analysis, embedding-based job ranking, and personalized application support, while implementing secure auth workflows with **token-based authentication** and **PBKDF2 password hashing**.

Mini Search Engine | C++, Open AI GPT-4

- Built a **C++ search engine** from scratch for large-scale information retrieval, implementing the full indexing and retrieval pipeline over **300K+** Wikipedia articles using **BM25 ranking**.
- Refactored into a **modular OOP architecture** (Tokenizer, InvertedIndex, Ranker, QueryProcessor), applying **SOLID principles** and **strategy patterns** to support pluggable ranking algorithms.
- Improved search relevance by adding **typo-tolerant query correction** and optional **GPT-4-powered query rewriting** with graceful fallback for offline reliability.
- Developed a modular backend architecture with **CMake-based build**, **benchmark tooling**, and **automated tests covering** for tokenization, ranking, indexing, and end-to-end query flows.

Distributed Task Queue System | Go, Redis

- Architected a **distributed task queue system** in **GO** using **Asynq, Redis, and Gin** to offload asynchronous email delivery and image-processing workloads from the synchronous request path.
- Designed a **multi-queue execution model** with **priority-based scheduling, delayed jobs, retries, timeouts, and dead-letter handling**, strengthening fault tolerance and enabling resilient processing of background workloads.
- Built operational tooling end to end, including queue inspection APIs, task lifecycle controls, structured observability, and an embedded monitoring dashboard, reducing production debugging time by **40%**.

TECHNICAL SKILLS

- **Language:** Python (NumPy), JavaScript/Typescript, C++ (STL, multi-threading), C# (.NET), SQL, Java, Go
- **Tools/Frameworks:** React, Node.js, Vite, JWT, Django, Docker, Kubernetes, Spring Boot, Express, Webpack, Jest, Git, CI/CD, CMake, FFmpeg, OpenGL, Google Test, Gin, Asynq
- **Systems/Databases:** MySQL, MongoDB, PostgreSQL, Redis, Microservices, AWS (Lambda, S3, API Gateway, DynamoDB), RESTful APIs, Message Queues, ElasticSearch